

# Predictive Analytics Explanation

## Customer Retention and Service Interaction Experience

### 1. Assignment Context and Predictive Analytics Goal

The selected analytical statement is:

**Customers who enjoy service interactions are more likely to return.**

The assignment asks the group to justify this statement using three types of analytics: descriptive analytics, inferential analytics, and predictive analytics. This document focuses only on the predictive analytics part.

#### 1.1 What predictive analytics means here

Predictive analytics means using past data to train a model that can predict an outcome for new data. In this assignment, the outcome is whether a customer is likely to return or be retained.

So the main predictive question is:

Can service interaction variables predict whether a customer will return?

The target variable used in both categories is `retention_label`. In the final binary modelling setup, the target is interpreted as:

| Label | Meaning                     |
|-------|-----------------------------|
| 0     | Not returned / Not retained |
| 1     | Returned / Retained         |

#### Simple explanation

The models are trained to learn patterns from customer service data. After learning from old records, the models can receive a new customer interaction and estimate whether that customer is likely to return.

### 2. Why Two Categories Were Used

The analytical statement is broad because “enjoy service interactions” can mean different things. To make the predictive analysis clearer, the work was divided into two categories. Each category used a separate dataset and a separate machine learning model.

| Category   | Focus                          | Main idea                                                                                                                       |
|------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Category 1 | Responsiveness & Support Speed | Checks whether fast responses and satisfaction-related support variables predict customer retention.                            |
| Category 2 | Interaction Experience         | Checks whether service quality, customer emotion, agent behaviour, helpfulness, and outcome quality predict customer retention. |

This two-category approach is useful because customer return behaviour may be affected by both the speed of support and the quality of the interaction. A customer may return because the support was fast, because the agent was polite and helpful, or because both happened together.

### 3. Overall Predictive Analytics Workflow

Both categories followed a similar machine learning workflow. This made the analysis consistent, professional, and easier to explain during the viva.

1. Load and understand the dataset.
2. Clean the dataset and prepare the target variable.
3. Perform exploratory data analysis to understand important patterns.
4. Check class imbalance in the target variable.
5. Split the data into training and testing sets using stratified sampling.
6. Build preprocessing pipelines for numeric, ordinal, nominal, and binary features.
7. Train baseline models and machine learning models.
8. Compare normal models, class-weighted models, and SMOTE-based models.
9. Evaluate models using accuracy, balanced accuracy, macro F1-score, precision, recall, confusion matrix, and ROC-AUC.
10. Perform hyperparameter tuning using GridSearchCV.
11. Select the final best model.
12. Analyze feature importance.
13. Export the final model pipeline and metadata for the Streamlit app.
14. Use weighted prediction to combine Category 1 and Category 2 probabilities.

#### Why this workflow is strong

The workflow does not only train one model and stop. It compares the model with baselines, handles class imbalance, tunes model settings, checks important features, and prepares the model for real use in a Streamlit application.

### 4. Key Concepts Used in Both Categories

| Concept            | Simple meaning                                                                                                                                |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Train-test split   | The dataset is divided into training data and testing data. The model learns from training data and is evaluated on unseen testing data.      |
| Stratified split   | Keeps the retained/not-retained class ratio similar in both training and testing sets. This is important because the datasets are imbalanced. |
| Pipeline           | Combines preprocessing and model training into one workflow. This reduces mistakes and helps prevent data leakage.                            |
| Baseline model     | A simple model used for comparison. If the real model cannot beat the baseline, it is not useful.                                             |
| Class imbalance    | Occurs when one target class has much more data than the other. Both datasets had more retained customers than not-retained customers.        |
| SMOTE              | Creates synthetic minority-class samples during training to help the model learn the smaller class better.                                    |
| Class weighting    | Gives more importance to the minority class during training.                                                                                  |
| GridSearchCV       | Tests different hyperparameter combinations using cross-validation and selects the best settings.                                             |
| Feature importance | Shows which input variables contributed most to the final prediction.                                                                         |

## 5. Category 1: Responsiveness & Support Speed

Category 1 checked whether support speed and service responsiveness can predict customer retention. This category represents the operational side of the customer service experience.

Main question: Do faster responses and higher satisfaction scores make customers more likely to return?

### 5.1 Dataset overview

| Item                           | Result |
|--------------------------------|--------|
| Original rows                  | 2,894  |
| Original columns               | 9      |
| Missing customer_remarks       | 1,915  |
| Missing response_minutes       | 48     |
| Missing retention_label        | 157    |
| Duplicate rows before cleaning | 246    |

The important columns in this dataset were:

- channel\_name: the support channel used by the customer.
- category and sub\_category: the type of issue or support request.
- tenure\_bucket: how long the agent had worked or their experience bucket.
- agent\_shift: the shift during which the support interaction happened.
- csat\_score: customer satisfaction score.
- response\_minutes: how long the customer waited for a response.
- retention\_label: the target variable showing whether the customer was retained or not.

### 5.2 Cleaning and preprocessing done

- customer\_remarks was removed because it had too many missing values and was raw text. The model focused on structured features.
- Rows with missing retention\_label were removed because supervised learning requires a known target value.
- csat\_score and response\_minutes were converted into numeric values.
- Missing numeric values were filled using the median.
- String columns were cleaned by stripping spaces.
- tenure\_bucket was ordinal encoded because it has a meaningful order.
- channel\_name, category, sub\_category, and agent\_shift were one-hot encoded because they are nominal categories.
- Numeric features were standardized so that the model treats numeric scales more fairly.

#### Why the full pipeline matters

In the final notebook, preprocessing was placed inside the machine learning pipeline. This is better than preprocessing the full dataset before splitting, because the model should not learn information from the test data before evaluation.

### 5.3 Target distribution and imbalance

| Class | Meaning                     | Count | Percentage |
|-------|-----------------------------|-------|------------|
| 1     | Returned / Retained         | 2,299 | About 84%  |
| 0     | Not returned / Not retained | 438   | About 16%  |

This shows that Category 1 is an imbalanced classification problem. Most customers belong to class 1. Because of this, a weak model could predict every customer as retained and still get high accuracy. Therefore, accuracy alone is not enough.

The analysis used better metrics such as balanced accuracy and macro F1-score because they treat both classes more fairly.

## 5.4 EDA output explanation

The EDA showed two very important patterns:

| Finding                                             | Explanation                                                                                                                                       |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| CSAT score strongly separated the classes.          | Low CSAT scores were mostly not-retained customers, while high CSAT scores were mostly retained customers.                                        |
| Response minutes were lower for retained customers. | The median response time for not-retained customers was about 16 minutes, while retained customers had a median response time of about 4 minutes. |

The CSAT output was especially strong:

| CSAT score | Not retained count | Retained count | Main interpretation                                     |
|------------|--------------------|----------------|---------------------------------------------------------|
| 1          | 393                | 0              | Very poor satisfaction was linked with not-retention.   |
| 2          | 37                 | 0              | Low satisfaction was linked with not-retention.         |
| 3          | 3                  | 0              | Middle score still mostly not-retained in this data.    |
| 4          | 5                  | 357            | High satisfaction was mostly retained.                  |
| 5          | 0                  | 1,942          | Excellent satisfaction was fully retained in this data. |

### EDA conclusion for Category 1

The EDA strongly supports the analytical statement. Customers with higher satisfaction and faster support responses were more likely to be retained.

## 5.5 Correlation heatmap output

| Feature          | Correlation with retention_label | Meaning                                                                     |
|------------------|----------------------------------|-----------------------------------------------------------------------------|
| csat_score       | About +0.96                      | Very strong positive relationship. Higher CSAT means higher retention.      |
| response_minutes | About -0.26                      | Negative relationship. Longer response time is linked with lower retention. |

The heatmap confirmed what the EDA already showed. CSAT score was the strongest feature related to retention. Response time also mattered, but its relationship was weaker than CSAT.

## 5.6 Models tested

- Most Frequent Class baseline.
- Stratified Random baseline.
- Logistic Regression.
- Logistic Regression with class weighting.
- Random Forest.
- Random Forest with class weighting.
- Logistic Regression with SMOTE.
- Random Forest with SMOTE.

## 5.7 Baseline model output

| Model               | Accuracy     | Balanced accuracy | Macro F1-score | Interpretation                                             |
|---------------------|--------------|-------------------|----------------|------------------------------------------------------------|
| Most Frequent Class | About 0.8394 | 0.5000            | About 0.4563   | High accuracy only because it predicts the majority class. |
| Stratified Random   | About 0.6971 | About 0.4612      | About 0.4626   | Random guessing based on class distribution; weak model.   |

The baseline result proves that accuracy alone is misleading. The most frequent baseline looked good by accuracy, but its balanced accuracy was only 0.50, meaning it did not properly identify both classes.

## 5.8 Machine learning model output

| Model group                        | Result                                                                                                     |
|------------------------------------|------------------------------------------------------------------------------------------------------------|
| Logistic Regression                | Perfect performance on the test set.                                                                       |
| Class-weighted Logistic Regression | Perfect performance on the test set.                                                                       |
| Random Forest                      | Perfect performance on the test set.                                                                       |
| Class-weighted Random Forest       | Perfect performance on the test set.                                                                       |
| SMOTE models                       | Also achieved perfect performance, but did not improve further because normal models were already perfect. |

The main model metrics were:

| Metric            | Result | Meaning                                                            |
|-------------------|--------|--------------------------------------------------------------------|
| Accuracy          | 1.0000 | All test predictions were correct.                                 |
| Balanced accuracy | 1.0000 | Both retained and not-retained classes were predicted correctly.   |
| Macro F1-score    | 1.0000 | The model performed perfectly across both classes.                 |
| ROC-AUC           | 1.0000 | The model perfectly separated retained and not-retained customers. |

### Important interpretation

The perfect result happened mainly because csat\_score almost completely separated the two target classes. This is strong evidence in the dataset, but it should be explained carefully in the viva.

## 5.9 SMOTE and hyperparameter tuning

SMOTE was tested because the minority class was only about 16%. However, SMOTE did not improve the result because the features already separated the classes very clearly.

Hyperparameter tuning was done using GridSearchCV. The tuned Logistic Regression still gave perfect performance, and it was easier to explain than Random Forest. Therefore, Logistic Regression can be treated as the most suitable final model for Category 1 because it is simple, interpretable, and highly accurate.

## 5.10 Feature importance output

| Feature                                                          | Importance interpretation                                                                     |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| csat_score                                                       | Most important predictor. It almost completely explained retention behaviour in this dataset. |
| response_minutes                                                 | Second important predictor. Faster response was linked with higher retention.                 |
| channel_name, category, sub_category, agent_shift, tenure_bucket | Low or near-zero importance compared to CSAT and response time.                               |

## 5.11 Model export and testing for Streamlit

After selecting the final model, the model was exported using joblib. The important decision was to export the full trained pipeline, not only the classifier.

This is correct because the full pipeline already includes:

- missing value handling,
- scaling,
- ordinal encoding,
- one-hot encoding,
- the final classifier.

Metadata was also saved with the model. This metadata included expected feature columns, dropdown options, class labels, package versions, and model name. This helps the Streamlit app create input fields and send the correct input format to the model.

The exported model was then loaded again and tested using sample manual inputs. This confirmed that the saved model file works correctly before using it in Streamlit.

## 5.12 Category 1 conclusion

The Category 1 results strongly support the analytical statement. Customers with higher CSAT scores and faster response times were more likely to be retained. The final model performed much better than the baseline models and achieved perfect test performance. The most important predictor was **csat\_score**, followed by **response\_minutes**.

## 6. Category 2: Interaction Experience

Category 2 checked whether the quality of the actual service interaction can predict customer retention. This category is directly connected to how the customer felt during the interaction and how well the issue was handled.

Main question: Do better interaction experience factors make customers more likely to return?

### 6.1 Dataset overview

| Item                           | Result |
|--------------------------------|--------|
| Original rows                  | 1,000  |
| Original columns               | 16     |
| Missing values                 | 0      |
| Duplicate rows before cleaning | 235    |

The important columns in this dataset were:

- `issue_category`: the type of issue handled.
- `customer_sentiment`: customer emotion during the interaction.
- `issue_complexity`: how complex the issue was.
- `agent_experience_level`: experience level of the agent.
- `agent_politeness`, `agent_empathy`, `agent_professionalism`: agent service behaviour indicators.
- `communication_clarity` and `helpfulness`: quality of explanation and support.
- `apology_present`, `patience_present`, `active_listening_present`, `deescalation_present`: binary behaviour indicators.
- `customer_tone_observed`: tone of the customer.
- `outcome_quality`: whether the issue was resolved or not.
- `retention_label`: target variable.

### 6.2 Target variable cleaning

| Original target label | Count |
|-----------------------|-------|
| retained              | 702   |
| uncertain             | 186   |
| not_retained          | 112   |

The **uncertain** rows were removed because the model was designed as a binary classification model. A binary model needs a clear yes/no target. Therefore, the final target became:

| Original label | Encoded value | Meaning                    |
|----------------|---------------|----------------------------|
| not_retained   | 0             | Customer was not retained. |
| retained       | 1             | Customer was retained.     |

| Class | Meaning      | Count | Percentage |
|-------|--------------|-------|------------|
| 1     | Retained     | 702   | 86.24%     |
| 0     | Not retained | 112   | 13.76%     |

This dataset was also imbalanced because most customers were retained. Therefore, the evaluation again focused on balanced accuracy, macro F1-score, and confusion matrix instead of only accuracy.

### 6.3 Cleaning and preprocessing done

- Text values were cleaned by removing spaces and converting them to lowercase.
- uncertain rows were removed to make the task binary.
- retained was mapped to 1 and not\_retained was mapped to 0.
- Binary columns were converted to integers.
- Ordinal columns were ordinal encoded because they have a meaningful order.
- Nominal columns were one-hot encoded because they do not have a natural order.
- Ordinal numeric outputs were standardized in the pipeline.

### 6.4 Feature groups used in Category 2

| Feature group    | Columns                                                                                                    | How they were handled                           |
|------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Binary features  | apology_present, patience_present, active_listening_present, deescalation_present                          | Converted to integer 0/1 and imputed if needed. |
| Ordinal features | issue_complexity, agent_experience_level, customer_sentiment, customer_tone_observed, outcome_quality      | Encoded using ordered categories, then scaled.  |
| Nominal features | issue_category, agent_politeness, agent_empathy, agent_professionalism, communication_clarity, helpfulness | One-hot encoded.                                |

#### Simple explanation

Ordinal features have a meaningful order. For example, unresolved is worse than partially\_resolved, and resolved is better than both. Nominal features do not have a clear order, so they were one-hot encoded instead.

### 6.5 EDA output explanation

The EDA showed that interaction quality variables were strongly connected with retention.

| EDA area                     | Main pattern                                                                                                                  | Conclusion                                                                              |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Customer sentiment           | Neutral and positive customers were mostly retained. Negative or frustrated customers had higher not-retained percentages.    | Customer emotion during the interaction matters.                                        |
| Customer tone observed       | Positive and neutral tone were more connected with retention. Negative tone had more not-retention.                           | Customer mood during the interaction is useful for prediction.                          |
| Outcome quality              | Resolved outcomes were mostly retained. Unresolved, partially resolved, and unknown outcomes had more not-retained customers. | Outcome quality was the clearest factor separating retained and not-retained customers. |
| Service behaviour indicators | Helpfulness, politeness, empathy, professionalism, and clarity supported retention when they were high/present.               | Better agent behaviour improves the chance of retention.                                |

#### EDA conclusion for Category 2

The strongest EDA message was that customers are much more likely to return when the issue is resolved and the service interaction is helpful, polite, clear, and positive.

### 6.6 Correlation heatmap output

The target-focused correlation heatmap showed that the strongest positive relationship with retention was outcome\_quality.

| Positive relationship with retention          | Meaning                                                     |
|-----------------------------------------------|-------------------------------------------------------------|
| outcome_quality                               | The better the outcome, the higher the chance of retention. |
| helpfulness_high                              | Highly helpful service is linked with retention.            |
| communication_clarity_high                    | Clear communication supports retention.                     |
| agent_politeness_high                         | Polite agents improve interaction experience.               |
| customer_sentiment and customer_tone_observed | More positive emotion and tone are linked with retention.   |
| agent_professionalism_high                    | Professional behaviour supports customer return.            |

## 6.7 Train-test split output

| Set          | Rows | Retained percentage | Not-retained percentage |
|--------------|------|---------------------|-------------------------|
| Training set | 651  | About 86.18%        | About 13.82%            |
| Testing set  | 163  | About 86.50%        | About 13.50%            |

The split used stratification, so the retained/not-retained ratio stayed similar in both the training and testing sets. This is important because the dataset is imbalanced.

## 6.8 Models tested

- Most Frequent Class baseline.
- Stratified Random baseline.
- Logistic Regression.
- Logistic Regression with class weighting.
- Random Forest.
- Random Forest with class weighting.
- Logistic Regression with SMOTE.
- Random Forest with SMOTE.

## 6.9 Baseline model output

| Model               | Accuracy | Balanced accuracy | Macro F1-score | Interpretation                                           |
|---------------------|----------|-------------------|----------------|----------------------------------------------------------|
| Most Frequent Class | 0.8650   | 0.5000            | 0.4638         | Looks good only because most customers are retained.     |
| Stratified Random   | 0.7607   | 0.4973            | 0.4973         | Random class-distribution-based prediction; weak result. |

Again, this shows why accuracy alone is dangerous. A simple model can get 86.5% accuracy by mostly predicting retained, but balanced accuracy shows it has not learned both classes properly.

## 6.10 Machine learning model comparison output

| Model                                | Accuracy | Balanced accuracy | Macro F1-score | ROC-AUC |
|--------------------------------------|----------|-------------------|----------------|---------|
| Random Forest + SMOTE                | 0.9939   | 0.9965            | 0.9871         | 0.9968  |
| Random Forest                        | 0.9877   | 0.9737            | 0.9737         | 0.9984  |
| Random Forest - Class Weighted       | 0.9816   | 0.9702            | 0.9613         | 0.9981  |
| Logistic Regression - Class Weighted | 0.9755   | 0.9858            | 0.9511         | 0.9971  |
| Logistic Regression                  | 0.9755   | 0.9283            | 0.9453         | 0.9981  |
| Logistic Regression + SMOTE          | 0.9571   | 0.9176            | 0.9098         | 0.9961  |

Before tuning, Random Forest + SMOTE was the best model. It predicted both classes very well and performed much better than the baseline models.

### 6.11 Best pre-tuning model confusion matrix interpretation

| Class        | Result                                                        |
|--------------|---------------------------------------------------------------|
| Not retained | 22 out of 22 not-retained customers were correctly predicted. |
| Retained     | 140 out of 141 retained customers were correctly predicted.   |
| Error        | Only 1 retained customer was incorrectly predicted.           |

This is a very strong result because the model did not ignore the minority class. It correctly identified all not-retained customers in the test set for that run.

### 6.12 Hyperparameter tuning output

Hyperparameter tuning was done for Logistic Regression and Random Forest using GridSearchCV. The goal was to find better model settings instead of depending only on default values.

| Tuned model               | Best CV Macro F1 | Important best parameters                                                                                |
|---------------------------|------------------|----------------------------------------------------------------------------------------------------------|
| Tuned Logistic Regression | About 0.9677     | C = 1, solver = liblinear, SMOTE k_neighbors = 3                                                         |
| Tuned Random Forest       | About 0.9635     | max_depth = None, min_samples_leaf = 1, min_samples_split = 5, n_estimators = 200, SMOTE k_neighbors = 3 |

### 6.13 Final best model output

After tuning and final comparison, the final best model was Tuned Random Forest.

| Metric            | Value  | Meaning                                                            |
|-------------------|--------|--------------------------------------------------------------------|
| Accuracy          | 0.9939 | Almost all predictions were correct.                               |
| Balanced accuracy | 0.9965 | The model performed extremely well on both classes.                |
| Macro F1-score    | 0.9871 | Very strong performance while treating both classes fairly.        |
| ROC-AUC           | 0.9987 | Excellent ability to separate retained and not-retained customers. |

### 6.14 Feature importance output

| Rank | Feature                | Importance interpretation                                                   |
|------|------------------------|-----------------------------------------------------------------------------|
| 1    | outcome_quality        | Strongest predictor. Whether the issue was resolved was the biggest factor. |
| 2    | issue_category         | Some issue types were more connected with retention behaviour.              |
| 3    | helpfulness            | Helpful service improved prediction of retention.                           |
| 4    | agent_politeness       | Politeness helped explain customer return behaviour.                        |
| 5    | customer_sentiment     | Customer emotion was useful for prediction.                                 |
| 6    | issue_complexity       | The difficulty of the issue affected retention prediction.                  |
| 7    | customer_tone_observed | Customer tone added useful signal.                                          |
| 8    | agent_experience_level | Experience level had some effect.                                           |
| 9    | agent_professionalism  | Professionalism contributed to prediction.                                  |
| 10   | deescalation_present   | De-escalation had smaller but still useful influence.                       |

#### Most important Category 2 conclusion

outcome\_quality was the strongest predictor. This means customers are much more likely to return when their problem is solved successfully.

## 6.15 Model export and testing for Streamlit

Category 2 should also be exported in the same way as Category 1. The full final pipeline should be saved with joblib, together with metadata.

The metadata should include:

- model name,
- target name,
- expected feature columns,
- binary feature names,
- ordinal feature names and category order,
- nominal feature names,
- dropdown options for Streamlit,
- class labels,
- package versions,
- test metrics if available.

After export, the model should be loaded again and tested using manual sample inputs. This confirms the model file is usable before connecting it to the Streamlit app.

## 6.16 Category 2 conclusion

The Category 2 results strongly support the analytical statement. The final Tuned Random Forest model achieved very high performance and clearly outperformed the baseline models. The most important predictor was outcome\_quality, followed by issue\_category, helpfulness, agent\_politeness, customer\_sentiment, issue\_complexity, and customer\_tone\_observed. This shows that interaction experience has strong predictive value for customer retention.

## 7. Comparison Between Category 1 and Category 2

| Area                               | Category 1: Responsiveness & Support Speed                                                          | Category 2: Interaction Experience                                          |
|------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Main focus                         | Speed of support and customer satisfaction.                                                         | Quality of interaction, agent behaviour, tone, sentiment, and outcome.      |
| Dataset size after binary cleaning | 2,737 rows approximately.                                                                           | 814 rows.                                                                   |
| Class imbalance                    | About 84% retained and 16% not retained.                                                            | About 86.24% retained and 13.76% not retained.                              |
| Best final model                   | Logistic Regression can be selected because it achieved perfect performance and is easy to explain. | Tuned Random Forest was the final best model.                               |
| Strongest feature                  | csat_score.                                                                                         | outcome_quality.                                                            |
| Secondary useful feature           | response_minutes.                                                                                   | helpfulness, issue_category, politeness, sentiment, tone.                   |
| Main conclusion                    | Higher satisfaction and faster support responses are linked with retention.                         | Resolved outcomes and better interaction quality are linked with retention. |

### Overall comparison conclusion

Both categories support the analytical statement, but they explain it from different sides. Category 1 shows that fast and satisfactory support matters. Category 2 shows that the actual quality and outcome of the interaction matter.

## 8. Weighted Prediction Concept for the Final Streamlit Application

The final Streamlit application can use both category models separately and then combine their prediction probabilities into one final prediction. This is called a weighted prediction approach.

### 8.1 Why combine both models?

The two models look at different parts of the customer service experience. Category 1 looks at speed and satisfaction. Category 2 looks at interaction quality and outcome. Combining them gives a more complete prediction than using only one category.

| Model                     | Output used for final combination                                                         |
|---------------------------|-------------------------------------------------------------------------------------------|
| Category 1 model          | Probability that the customer will be retained based on responsiveness and support speed. |
| Category 2 model          | Probability that the customer will be retained based on interaction experience.           |
| Final weighted prediction | Combined probability using selected category weights.                                     |

### 8.2 Why Option 3 was selected

Option 3 is the topic-based weighting approach. Instead of giving equal weights only because it is simple, the weights are chosen based on how directly each category represents the analytical statement.

| Category                                   | Weight | Reason                                                                                                                                                                        |
|--------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Category 1: Responsiveness & Support Speed | 40%    | Important because speed and satisfaction affect customer experience.                                                                                                          |
| Category 2: Interaction Experience         | 60%    | Given higher weight because it directly measures customer enjoyment through politeness, empathy, professionalism, clarity, helpfulness, sentiment, tone, and outcome quality. |

### Why this is acceptable

The assignment statement is about customers who enjoy service interactions. Category 2 directly measures interaction enjoyment and outcome quality, so giving it a slightly higher weight is logically justified. Category 1 still matters because support speed and satisfaction also affect the overall service experience.

## 8.3 Weighted prediction formula

The final probability can be calculated as:

$$\text{Final Retention Probability} = (\text{Category 1 Probability} \times 0.40) + (\text{Category 2 Probability} \times 0.60)$$

Then the final class can be selected using a threshold:

If Final Retention Probability  $\geq 0.50 \rightarrow$  Predict Retained

If Final Retention Probability  $< 0.50 \rightarrow$  Predict Not Retained

## 8.4 Example weighted prediction

| Step                            | Value                                             |
|---------------------------------|---------------------------------------------------|
| Category 1 retained probability | 0.82                                              |
| Category 2 retained probability | 0.74                                              |
| Category 1 weight               | 0.40                                              |
| Category 2 weight               | 0.60                                              |
| Final probability               | $(0.82 \times 0.40) + (0.74 \times 0.60) = 0.772$ |
| Final prediction                | Retained, because 0.772 is greater than 0.50.     |

## 8.5 Streamlit application flow

- Load the Category 1 model bundle and metadata.
- Load the Category 2 model bundle and metadata.
- Display input fields for Category 1 variables.
- Display input fields for Category 2 variables.
- Use the Category 1 model to predict retained probability.
- Use the Category 2 model to predict retained probability.
- Apply the weighted prediction formula using 40% Category 1 and 60% Category 2.
- Display each category prediction separately.
- Display the final combined retention probability and final prediction label.

## 8.6 Limitations of weighted prediction

- The two models were trained on separate datasets, so the final weighted probability is an application-level combination, not a single unified model trained on combined customer records.
- The weights are logically justified, but they are not mathematically optimized unless tested on a common validation dataset.
- If possible in the future, a combined dataset with both Category 1 and Category 2 variables for the same customers would allow training one unified model.
- Because Category 1 achieved perfect performance mainly due to CSAT, the result should be interpreted carefully if retention\_label was derived from CSAT.
- Model probabilities may need calibration in a production-level system, especially when probabilities are combined.

## 9. Viva-Ready Questions and Answers

| Question                                                         | Short answer                                                                                                                                                                      |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What is the main aim of predictive analytics in this assignment? | To predict whether a customer will return or be retained using service interaction variables.                                                                                     |
| Why did you use two categories?                                  | Because customer service experience has more than one dimension. Category 1 measures responsiveness and support speed, while Category 2 measures interaction quality and outcome. |
| What is the target variable?                                     | retention_label. It was encoded as 1 for retained/returned and 0 for not retained/not returned.                                                                                   |
| Why were uncertain rows removed in Category 2?                   | Because the model was binary. uncertain is not clearly retained or not retained, so it was removed to create a clear supervised learning task.                                    |
| Why did you use pipelines?                                       | Pipelines combine preprocessing and modelling into one workflow. They reduce mistakes and help avoid data leakage.                                                                |
| What is data leakage?                                            | Data leakage happens when information from the test data accidentally influences training. This can make results look better than they really are.                                |
| Why was stratified train-test split used?                        | Because the classes were imbalanced. Stratification keeps the same class ratio in training and testing sets.                                                                      |
| Why is accuracy not enough?                                      | Because both datasets had many more retained customers. A model could predict retained for everyone and still get high accuracy.                                                  |
| Which metrics were more important?                               | Balanced accuracy, macro F1-score, confusion matrix, recall, precision, and ROC-AUC.                                                                                              |
| Why was SMOTE used?                                              | Because the minority class was small. SMOTE creates synthetic minority-class samples during training so the model can learn the smaller class better.                             |
| What was the best model in Category 1?                           | Logistic Regression can be selected because it achieved perfect performance and is simple to explain.                                                                             |
| What was the strongest feature in Category 1?                    | csat_score was the strongest predictor, followed by response_minutes.                                                                                                             |
| What was the best model in Category 2?                           | Tuned Random Forest was the final best model.                                                                                                                                     |
| What was the strongest feature in Category 2?                    | outcome_quality was the strongest predictor.                                                                                                                                      |
| Why did Category 1 achieve perfect performance?                  | Because csat_score almost completely separated retained and not-retained customers.                                                                                               |
| Is perfect accuracy always good?                                 | It is strong, but it must be interpreted carefully. If the target label was created directly from CSAT, the model may be learning the label creation rule.                        |
| Why export the full pipeline?                                    | Because the pipeline contains preprocessing steps and the classifier. Exporting only the classifier would not handle raw Streamlit inputs correctly.                              |
| What is weighted prediction?                                     | It combines the retained probabilities from both category models using selected weights.                                                                                          |
| Why use 40% and 60% weights?                                     | Category 2 gets 60% because it directly measures customer enjoyment and interaction quality. Category 1 gets 40% because responsiveness and support speed are also important.     |
| What is the final formula?                                       | Final probability = Category 1 probability $\times$ 0.40 + Category 2 probability $\times$ 0.60.                                                                                  |

## 10. Final Overall Conclusion

The predictive analytics part of the assignment strongly supports the analytical statement that customers who enjoy service interactions are more likely to return.

**Category 1** showed that customers with higher CSAT scores and faster response times were more likely to be retained. The strongest feature was `csat_score`, followed by `response_minutes`. The machine learning models performed much better than the baselines, and Logistic Regression can be selected as the final model because it achieved perfect performance while remaining simple and explainable.

**Category 2** showed that interaction experience variables also strongly predicted retention. The final best model was Tuned Random Forest, and the strongest feature was `outcome_quality`. This means customers are more likely to return when their issue is resolved successfully and when the interaction is helpful, polite, clear, and professional.

Finally, the Streamlit application can combine both model outputs using a topic-based weighted prediction. Category 1 contributes 40% and Category 2 contributes 60%. This gives a final retention probability that considers both support speed and interaction quality. This weighted approach is suitable because the analytical statement focuses mainly on customer enjoyment during service interactions, while still recognizing that responsiveness and support speed also affect customer return behaviour.